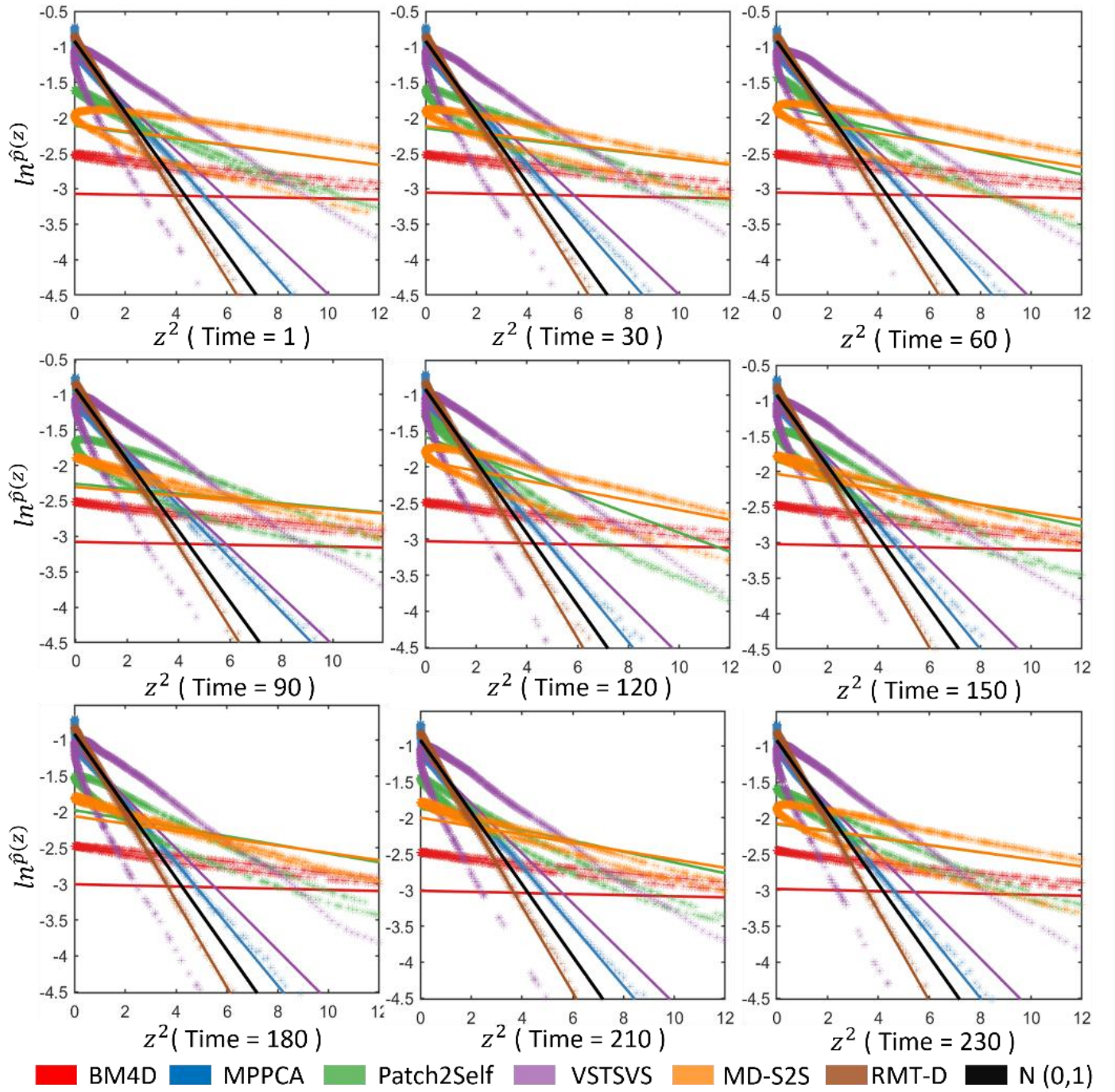




Distributional diagnostics for standardized residuals within the brain foreground in the acquired human rs-fMRI data. Markers show empirical log-density estimates $\ln \hat{p}(z)$ plotted against z^2 , obtained using Gaussian kernel density estimation from finite standardized residuals within the common valid brain foreground. Solid curves show the corresponding zero-mean Gaussian log-density curves, with the standard deviation estimated from all finite standardized residuals for each method. The display range is truncated only for visualization and does not affect the estimation of the standard deviation. All methods use the noise scale estimated by RMT-D as the common normalization reference. The black curve represents the standard Gaussian distribution $\mathcal{N}(0,1)$, which is included only as a descriptive reference.